

# Symmetry as a Design Axis in Variational Quantum Learning

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**Abstract**—The success of variational quantum learning models crucially depends on choosing parametrizations that reflect the structure of the problem at hand. Symmetries provide one of the clearest such structures: whenever transformations of the input leave the desired outcome unchanged, this invariance should be built into the model rather than discovered during training. However, imposing a symmetry does not by itself determine a useful ansatz. Even within the symmetry-preserving space, one still has to decide where the trainable degrees of freedom should be placed. In this work, we study this remaining design freedom in equivariant variational quantum circuits. Building on symmetry-based parameter sharing, we disentangle two architectural choices that are often conflated: how much of the available symmetry should be enforced, and which symmetry-respecting interactions should be made trainable. Using Tic-Tac-Toe as a fully enumerable and structurally transparent test case, we find that relaxing full symmetry to suitable subgroups preserves most of the generalization benefit. By contrast, the dominant performance gains arise from gates that act directly on the decisive motifs of the task. Our results suggest a general principle for quantum ansatz design: symmetry is a powerful way to restrict the search space, but the effectiveness of the model is determined by placing trainable interactions where the structure of the problem actually lives.

**Index Terms**—quantum machine learning, equivariance, symmetry, inductive bias, variational quantum circuits, parameter sharing, generalization

## I. INTRODUCTION

A central challenge in variational quantum machine learning is not merely to build expressive quantum circuits, but to build the right expressive circuits. Layered parameterized circuits with data re-uploading [1], [2] can be highly flexible, but flexibility alone is not a design principle. Without architectural guidance, a model may spend parameters and data on distinctions irrelevant to the task, while failing to allocate capacity to the structures that determine the label. The practical question is therefore not only how much expressivity a quantum model has, but where this expressivity is placed.

Symmetry is one of the few sources of prior knowledge that can be stated exactly. Consider a target map  $y : \mathcal{X} \rightarrow \mathcal{Y}$  and a group  $S$  acting on the input space through transformations  $V_s : \mathcal{X} \rightarrow \mathcal{X}$ . The task is invariant under  $S$  if

$$y(V_s[x]) = y(x) \quad \text{for all } x \in \mathcal{X}, s \in S.$$

In words, a symmetry specifies transformations of the input that must not change the correct prediction. A translated image still depicts the same object, a permuted graph still represents the same graph, and a rotated Tic-Tac-Toe board still has the same winner. A model that respects this structure by construction need not rediscover it from data.

This observation is central to geometric deep learning: classical architectures such as convolutional neural networks organize trainable degrees of freedom around task symmetries [3]. In quantum machine learning, geometric quantum machine learning represents known transformations on the Hilbert space and constrains circuits accordingly [4], [5]. Related work has developed group-invariant models, equivariant quantum neural networks, permutation-equivariant architectures, graph-equivariant circuits, and symmetry-informed variational ansätze [4]–[7].

Meyer et al. [8] gave a concrete blueprint for this program. If a data embedding  $U(x)$  satisfies

$$U(V_s[x]) = U_s U(x) U_s^\dagger,$$

then the data symmetry is represented by a unitary action  $U_s$ . Trainable blocks that commute with this representation, together with compatible state preparation and readout, yield a model whose prediction is invariant by architecture rather than by optimization.

However, symmetry alone does not specify an ansatz. It tells us which inputs should be identified, which parameters may be shared, and which observables should be aggregated, but not where trainable interactions should live. Two circuits can respect the same group action and still match the task very differently. We therefore treat symmetry not as a switch, but as a constrained design space in which one must still choose the sharing strength and the motifs on which trainable quantum interactions are placed.

Tic-Tac-Toe is a clean laboratory for this question. The task is finite, exactly enumerable, and structurally transparent: each board has nine fields and the label is determined by explicit three-cell winning lines. The relevant symmetry is the dihedral group  $D_4$ , containing the eight rotations and reflections of the square. Acting on a  $3 \times 3$  board,  $D_4$  permutes the nine fields while preserving the corner, side, and center orbits. Since these transformations do not change whether crosses win, circles win, or the game is a draw, the classification task is  $D_4$ -invariant.

Each board value  $g_i \in \{-1, 0, +1\}$  is encoded on one qubit through a Pauli- $X$  rotation with angle  $x_i = 2\pi g_i/3$ . Under a board symmetry, the encoded qubits are permuted like the board fields, so equivariance can be imposed by sharing parameters across symmetry orbits of gates.

We separate two design choices that are usually entangled. The first is how much symmetry to impose: full  $D_4$ -equivariance is natural, but smaller subgroups may preserve most of the useful bias while leaving the circuit more flexible.

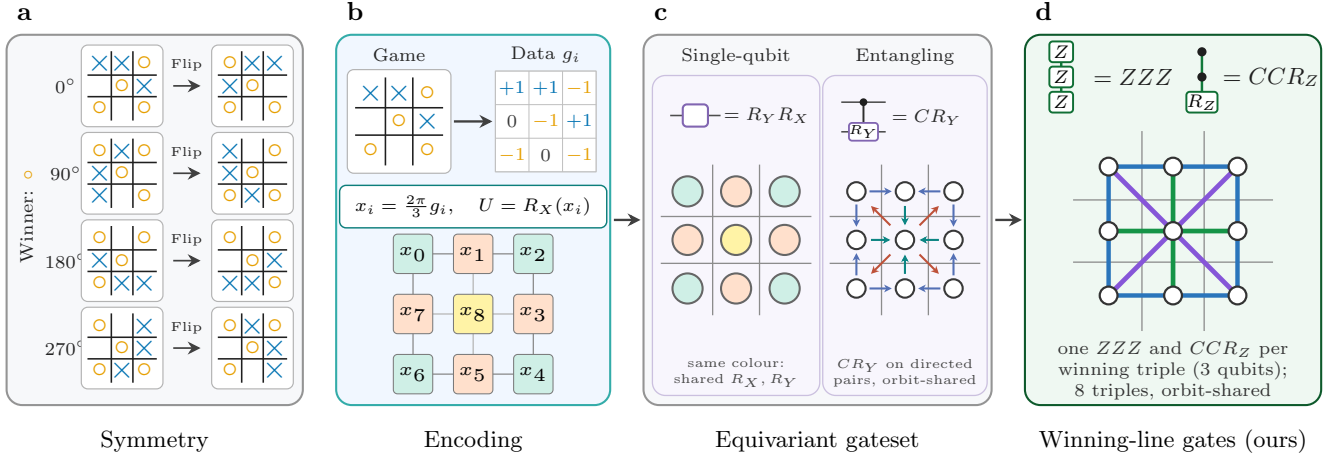


Fig. 1. From task symmetry to a task-aligned equivariant circuit. (a) Tic-Tac-Toe labels are invariant under the  $D_4$  rotations and reflections of the board. (b) Board entries  $g_i \in \{-1, 0, +1\}$  are encoded as  $R_X(x_i)$  rotations with  $x_i = \frac{2\pi}{3}g_i$ ; the chosen indexing exposes the corner, side, and center orbits. (c) The Meyer et al. [8] edge ansatz uses orbit-shared  $R_Y R_X$  rotations and directed  $C R_Y$  entanglers on neighbouring and center-connected fields. (d) Our extension adds orbit-shared  $ZZZ$  and symmetric  $CCR_Z$  gates on the eight winning triples, directly matching the label-relevant motifs while preserving equivariance. Panel layout adapted from Meyer et al. [8].

The second is where trainable interactions are placed. The canonical  $D_4$ -equivariant Tic-Tac-Toe ansatz of Meyer et al. uses orbit-shared single-qubit rotations and directed controlled rotations on neighboring or center-connected fields [8]. This respects the board symmetry, but the label is defined by complete rows, columns, and diagonals. We therefore add task-aligned, orbit-shared  $ZZZ$  and symmetric controlled-controlled- $R_Z$  interactions on these winning triples.

Our numerical results support this design-oriented view. Enforcing board symmetry improves generalization over an unshared edge ansatz, and suitable subgroups recover much of the performance of full  $D_4$ -equivariance. The largest improvement, however, comes from adding winning-line interactions to the equivariant circuit. Thus, symmetry is not merely a constraint that makes a model smaller. It is a language for organizing circuit design, but the effectiveness of the model still depends on placing trainable degrees of freedom where the structure of the task actually lives.

## II. PROTOCOL

Each legal, reachable Tic-Tac-Toe board is represented by nine values  $g_i \in \{-1, 0, +1\}$  for circle, empty, and cross. The three labels are circle win, draw, and cross win, encoded as  $(+1, -1, -1)$ ,  $(-1, +1, -1)$ , and  $(-1, -1, +1)$ . As in Meyer et al. [8], unfinished games are counted as draws. Since the full set contains only 5478 legal, reachable boards, we enumerate it exactly and draw class-balanced train and test splits from this set. The labels are invariant under the eight rotations and reflections of the board, forming the dihedral group  $D_4$ ; this symmetry is illustrated in Fig. 1(a).

We use one qubit per field, as shown in Fig. 1(b). Each value  $g_i$  is mapped to the angle  $x_i = 2\pi g_i/3$  and encoded by

an  $R_X$  rotation,

$$U(g) = \bigotimes_{i=0}^8 R_X\left(\frac{2\pi}{3}g_i\right).$$

The fields are numbered cyclically around the rim, with the center last. This makes the three  $D_4$  position orbits explicit: corners  $\{0, 2, 4, 6\}$ , sides  $\{1, 3, 5, 7\}$ , and center  $\{8\}$ .

We separate the imposed symmetry from the choice of gates. Let  $H \subseteq D_4$  be the subgroup that we enforce. For  $h \in H$ , let  $V_h$  act on the board and let  $P_h$  be the corresponding qubit permutation. The encoding is equivariant: transforming the board before encoding is equivalent to encoding first and then permuting the qubits,

$$U(V_h g) = P_h U(g) P_h^\dagger.$$

A trainable block  $W_H(\theta)$  preserves this symmetry if

$$P_h W_H(\theta) P_h^\dagger = W_H(\theta) \quad \text{for all } h \in H.$$

In practice, we enforce this by sharing parameters across gates whose supports lie in the same  $H$ -orbit. For directed gates, the order of the support is kept fixed, so control and target roles are not accidentally exchanged. Thus the symmetry fixes the sharing pattern, but not the gate family itself.

The circuit uses repeated data re-uploading layers. Each layer applies the encoding  $U(g)$  followed by  $p$  repetitions of a trainable block. Unless stated otherwise, we use  $L = 3$  layers and  $p = 2$  repetitions, keeping depth fixed across all sweeps.

Our baseline is the edge ansatz of Meyer et al., shown in Fig. 1(c). It applies local  $R_Y R_X$  rotations and directed controlled- $R_Y$  entanglers on neighbouring rim fields and selected center connections. Symmetry enters only through

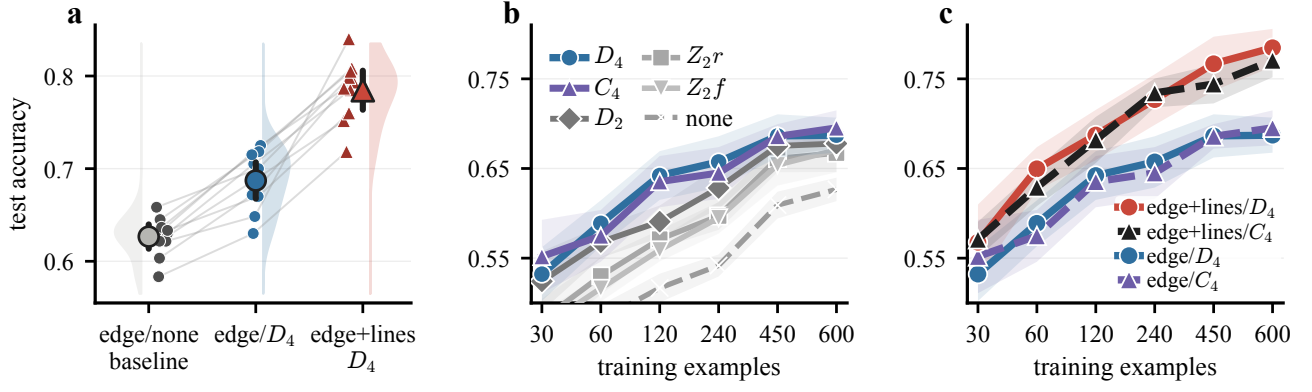


Fig. 2. Main empirical evidence. (a) Full  $D_4$  equivariance improves the original edge ansatz over no sharing. (b) The subgroup sweep across training sizes shows that  $C_4$  tracks  $D_4$  closely. (c) Adding winning-line interactions to the equivariant edge ansatz improves test accuracy while preserving the imposed symmetry.

parameter sharing: gates in the same  $H$ -orbit share the same parameter or parameter tuple.

The readout averages Pauli- $Z$  expectations over the three position orbits,

$$r_O(g) = \frac{1}{|O|} \sum_{i \in O} \langle Z_i \rangle_g,$$

for corners, sides, and center. These three values are used as class scores. Since the readout orbits are invariant under all subgroups considered here, the readout is compatible with every symmetry sweep.

This gives our first design axis: the amount of symmetry imposed. We keep encoding, edge wiring, gate family, depth, and readout fixed, and vary only the subgroup  $H$ . With  $r$  denoting a  $90^\circ$  rotation and  $f$  a reflection, we compare  $\{e\}$ ,  $\langle r^2 \rangle$ ,  $\langle f \rangle$ ,  $\langle r \rangle = C_4$ ,  $\langle r^2, f \rangle = D_2$ , and  $\langle r, f \rangle = D_4$ . For the edge ansatz at  $L = 3, p = 2$ , these variants have 204, 108, 126, 60, 78, and 54 trainable parameters, respectively.

The second design axis is the interaction structure. The edge ansatz couples neighbouring fields, but Tic-Tac-Toe labels are decided by complete three-cell winning lines. We therefore add orbit-shared gates on the eight winning triples: three rows, three columns, and two diagonals, as shown in Fig. 1(d). Each triple receives one  $ZZZ$  rotation and one  $CCR_Z$  interaction. The resulting edge+lines model keeps the original edge ansatz but adds trainable interactions directly on the label-relevant motifs. At  $L = 3, p = 2$ , the full- $D_4$  edge model has 54 parameters, while the full- $D_4$  edge+lines model has 90.

### III. RESULTS

This section evaluates the two design axes introduced above: the strength of imposed symmetry and the choice of trainable interactions. We first compare no sharing, partial subgroup sharing, and full  $D_4$  sharing with the edge circuit fixed. We then add winning-line interactions and use controls to separate symmetry-aware sharing from mere parameter reduction. All runs use mean-squared error, Adam with learning rate 0.01 [9], batch size 15, 30 minibatch updates per epoch, 100 epochs,

fixed seed blocks, PennyLane simulation [10], and PyTorch optimization [11].

Across these experiments, three trends emerge, summarized in Fig. 2. First, the unconstrained edge ansatz has more parameters but generalizes worse than the  $D_4$ -equivariant edge model. At 600 training examples, edge/none reaches about 0.63 test accuracy, while edge/ $D_4$  reaches about 0.69 with a smaller train-test gap. This supports the basic equivariance effect: the correct board symmetry improves generalization by shaping the hypothesis class.

Second, the symmetry effect is not strictly all-or-nothing. Full  $D_4$  remains the natural symmetry choice, but  $C_4$  rotation sharing tracks it closely across the data grid and is slightly ahead at the smallest training size. We interpret this as evidence that symmetry strength is a tunable architectural bias: partial symmetry can preserve most of the benefit while leaving the circuit less constrained. The random-sharing control in Fig. 3(a) confirms that the gain is not caused by parameter reduction alone; parameter-matched sharing without group-orbit structure performs worse.

The largest improvement comes from changing the interaction structure. Adding winning-line gates to the equivariant edge circuit, as shown in Fig. 2(c), raises the best  $D_4$  model from about 0.69 to roughly 0.80 test accuracy, with the  $C_4$  variant close behind. The edge ansatz can only represent rows, columns, and diagonals indirectly, whereas the line gates act directly on the motifs that determine the label. This is the central result: symmetry helps, but task-aligned equivariant interactions help more.

The controls in Fig. 3 rule out simpler explanations. The effect is not arbitrary parameter reduction, because random sharing underperforms orbit sharing. It is also not just parameter count: edge+lines/ $D_4$  has 90 parameters, far fewer than the 204-parameter unshared edge baseline, yet performs better. The ablations show that line channels help on their own, but the strongest model combines edge interactions with both winning-line channels.

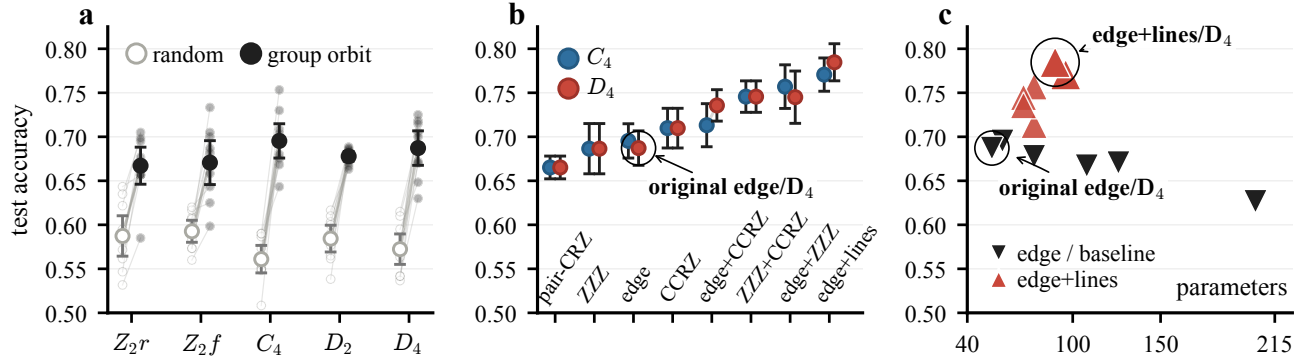


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing. (b) Ablations are anchored by the labelled original edge/ $D_4$  ansatz; the strongest model combines edge interactions with both line channels. (c) Accuracy is not explained by parameter count alone.

#### IV. DISCUSSION

Our results support a design-oriented view of equivariance in variational quantum learning: symmetry is a powerful inductive bias, but not a complete ansatz specification. It determines how parameters should be shared and how symmetry-related inputs should be treated, yet it does not decide which trainable interactions the circuit should contain. The subgroup sweep shows that equivariance is therefore not a binary choice. Full  $D_4$  remains the principled symmetry of Tic-Tac-Toe, but  $C_4$  recovers nearly the same generalization behavior, suggesting that partial symmetry can preserve the main inductive bias while avoiding unnecessary constraints. The largest gains instead come from the interaction structure. The original edge ansatz must represent winning lines indirectly, whereas our line-based gates act directly on the three-cell motifs that define the label. Together with the random-sharing and parameter-count controls, this shows that performance is driven not by fewer parameters alone, but by placing trainable degrees of freedom where the task geometry requires them. Tic-Tac-Toe is small and exactly enumerable, but precisely for this reason it isolates a general ansatz-design principle: impose the known symmetry, then choose gates that reach the task-relevant motifs. Future work should test softer relaxations of this recipe, for example shared angles with regularized gate-specific deviations, to interpolate between strict equivariance and fully free circuits.

#### APPENDIX

**Data generation and sampling.** Following Meyer et al. [8], we enumerate the 5478 legal, reachable boards by expanding the game tree and stopping at terminal states; unfinished positions count as draws. Splits are class-balanced; the 600-board test set is fixed and shared across models.

**Reproducibility.** The accompanying [anonymous review repository](#) contains the code, checked result tables, figure-generation scripts, and manuscript sources needed to reproduce the reported results, plots, and short paper. This makes the full workflow, from experiment outputs to the final PDF, directly auditable and reproducible.

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